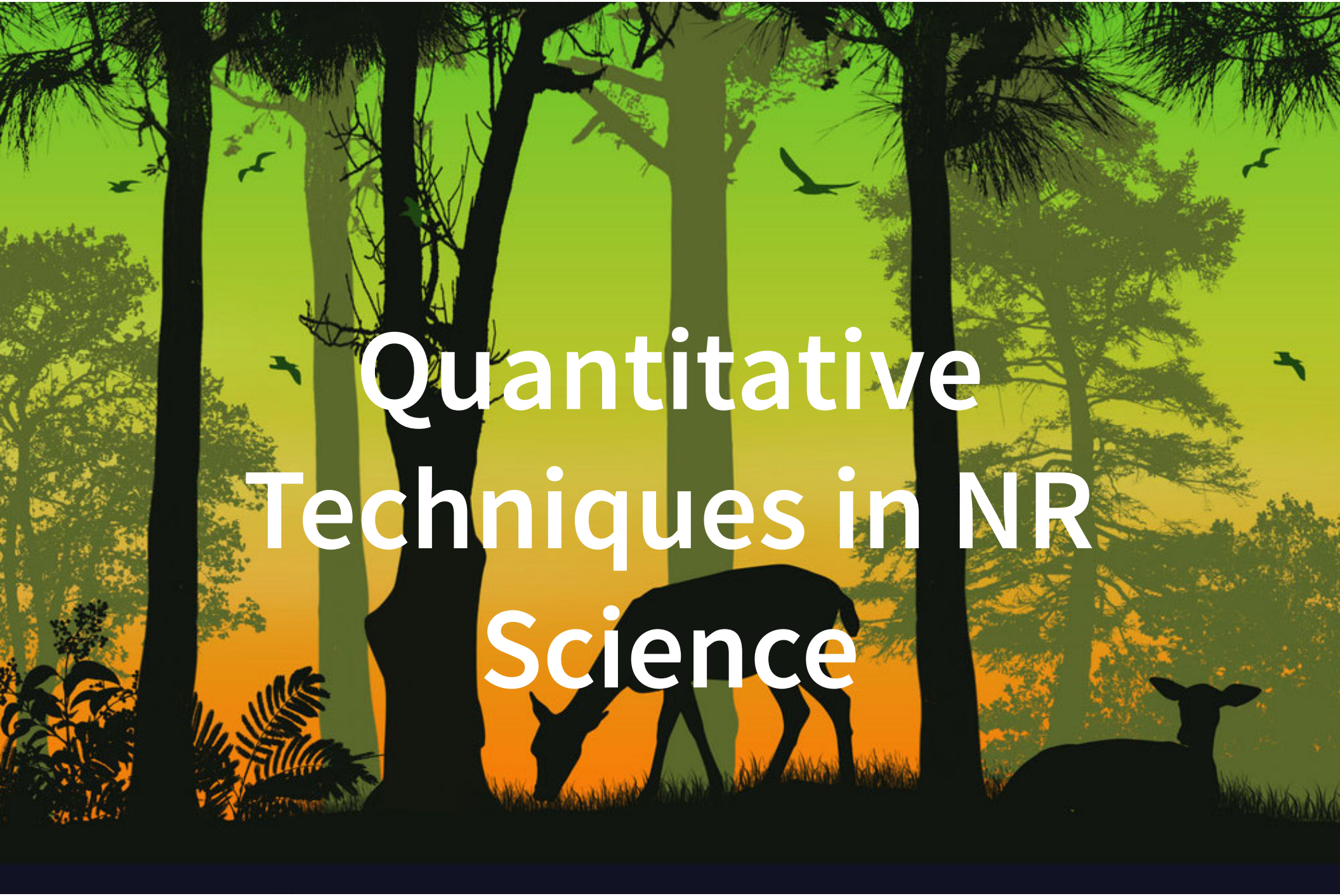


Quantitative Techniques in NR Science

Class Stuff

- **Me:** Dr. Brian Gerber
- **Class**: CI 117A, Monday 2pm-4:45pm
  - Lecture/Lab/Discussion
- **My Office**: CI 109
  - No set office hours
  - bgerber@uri.edu
- **Computers**: Always bring a laptop to class
  - Software/R Packages should be installed prior to lab (posted on main website)

What is this course?

Quantitative Techniques in Natural Resources Science

- A work in progress

¯_(ツ)_/¯

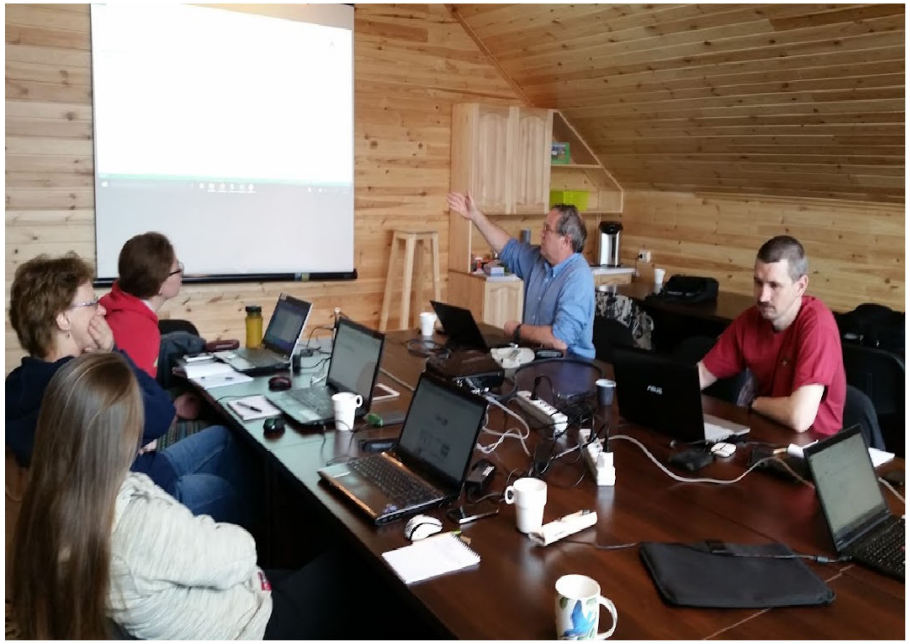
- Adaptive!
- Scope/depth depends on current baseline knowledge
- mixture of stats/modeling/code/math/science philosophy/ecology

Our Goals

- Better understand statistical modeling
- Learn to test assumptions in design and modeling
- Principles of statistical inference
- Quantitative literacy / Statistical Notation
- Science philosophy
- Coding reproducibility and share-ability
- Knowledge to carry across projects
  - not highly specialized statistical models

- Able to read modern ecological literature
- Understand what you are doing when using data and models; **coding/statistics/modeling are related but not the same**
- Coding skills (highly marketable)
- Taking control of evaluating assumptions
- Collaborative with statisticians
- Approach data and data analysis with an appropriate

Who am I?



And you…

- What is your name?
- What is your program/project/field of study?
- Why do you want to learn more about ‘quantitative techniques’?
- What is your first emotion when you hear **statistics**, **math**, or **coding**?

Science Philosophy

I am a pragmatist that believes theoretical foundations and logical justification are critical to science and scientific learning for decision making.

Teaching Philosophy

- I can not make you learn; learning is a choice (in every movement)
- An inclusive environment is paramount for learning
- Communication is key
- Everyone has something to teach and something to learn
- Struggle is good. Solving problems leads to learning
- BUT….

Tentative Schedule

https://bgerber123.github.io/Schedule/Schedule.html

Grades

- Attendance and Participation (10%)
  - let me know prior to class if you can not make it
- Weekly Assignments (60%)
  - Due the following class
- Project (30%)

Project

Groups (2-3) will design a

reading/discussion - lecture - lab - HW

around a statistical / methodology topic relevant to ecology and resource mgmt.

- work with me
- the topic needs to be general(ish)
- lead one class period

R Code Language

Write down, what is the name of the

- object?
- function?
- argument(s)?

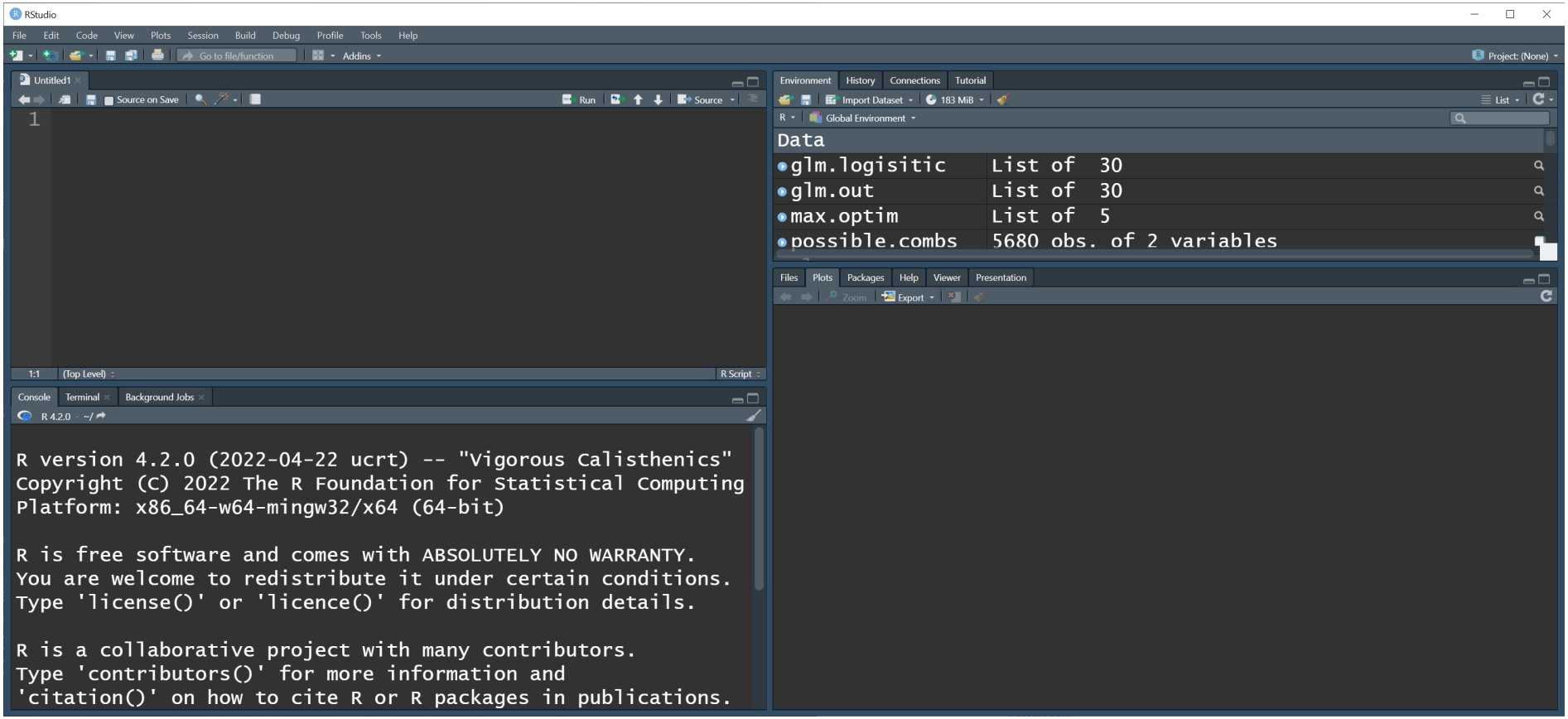
```
a = seq(0,1,by=0.1)
```

Note: code can be copied to your clipboard (top-right).

Code scripts will also be on the website.

RStudio

What does each panel do?



Importance of your computer's CPU and memory?

Getting help with Code

- Share via RStudio Server (ideally)
- If emailing, include a script file and RData file (workspace)
- Hierarchical code organization
  - code structure using indenting
  - top –> bottom execution
  - remove all extraneous code (minimal working example)
  - the code you send is likely different than what you are working on

Help! My for loop doesn't work…

```
                    cor.sp.route.cor=vector("list",n.speci
cor.sp=rep(NA,n.species)
                for(s in 1:n.species){
route=new.cov.species.long.scaled[[s]]$routeID
cor.sp[s]=cor(patch.size20.species.scaled.mat.center.r
    for(i in 1:nroutes){
temp1=patch.size20.species.scaled.mat.center.route[s,w
  temp2=patch.count20.species.scaled.mat.center.route[
  if(length(temp1)>5){
  cor.sp.route.cor[[s]]=abs(c(cor.sp.route.cor[[s]],co
  }}}
```

Minimal/Annotated working example

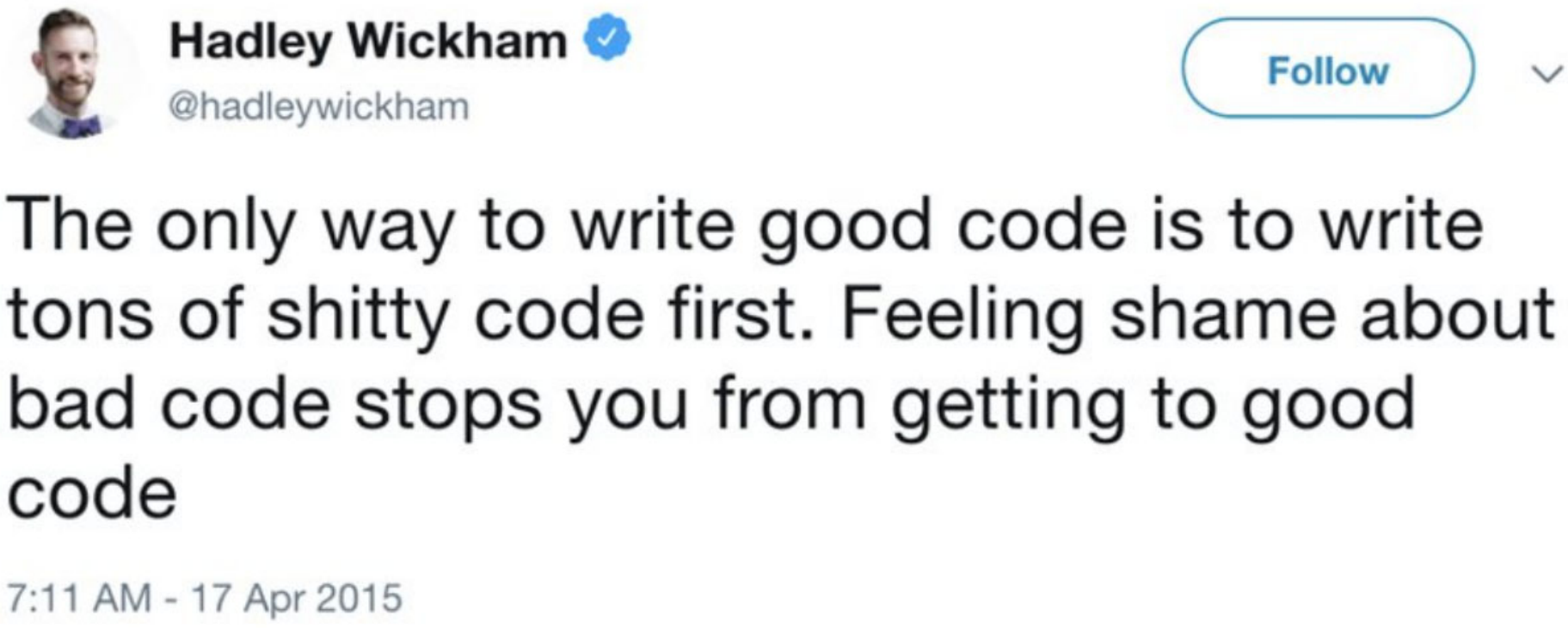
```
#Load data (file provided)
  cov.core.20k=load("habitat_radius_20k_core.RData")

#Create a list
  cor.sp.route.cor=vector("list",12)

```

```
#Double loop with if statement
  for(s in 1:cor.sp.route.cor){
    cor.sp[s]=cor(cor.sp.route.cor[s,],cor.sp.route.co
      for(i in 1:length(cor.sp.route.cor)){
          if(length(cor.sp.route.cor[i,])>5){
            blah blah blah
          } #End if statement
```

Code


Hadley Wickham
@hadleywickham
Follow
The only way to write good code is to write tons of shitty code first. Feeling shame about bad code stops you from getting to good code
7:11 AM - 17 Apr 2015